

Dimitri Chrysafis

GPU systems, real-time graphics, geometry, and ML

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CS at UW–Madison (**GPA 4.0**). I build GPU-accelerated simulation, graphics, and geometry tools end-to-end—kernels and math first, then demos you can open in a browser. Open-source contributor; also a competitive marathon and triathlon athlete.

Education

University of Wisconsin–Madison / B.S. Computer Science

Expected May 2027

Skills

GPU / graphics systems (WebGPU, Metal, CUDA/MPS, WGSL/GLSL), geometry pipelines, PyTorch.

Open Source

SimpleX Chat / [privacy-first messaging](#)

Contributor

- Worked on multi-device sync in the client pipeline—followed message and state flow across linked devices and fixed reconciliation edge cases that showed up in real use.
- Tightened connection lifecycle handling so state stays coherent through disconnects, reconnects, and concurrent sync (fewer duplicate-delivery paths).

Selected Projects

Interactive 3D Fluid Simulator

[demo](#) | [code](#) | WebGPU, WGSL

- MLS-MPM in the browser: **400k particles** in real time with parallel particle↔grid compute passes.
- No CPU particle round-trips—shared storage buffers, direct draw, 128-thread workgroups, atomic fixed-point accumulation, piston boundaries, turbulence, adaptive density.

Volumetric Sphere Packing

[code](#) | [write-up](#) | PyTorch, Open3D

- Non-overlapping tangent spheres inside arbitrary OBJ meshes (Möller–Trumbore parity, nearest-surface queries, grid seeding, stochastic growth).
- Containment / clearance / overlap as batched CUDA/MPS ops; binary-search radius fit, decimation, surface sampling, live Open3D view.

Tennis Match Prediction

CatBoost, Python

- CatBoost on **92k ATP matches (1982–2024)** with surface Elo, H2H, form, and fatigue from tournament schedules.
- **83.7% accuracy** (~12% over baselines) after iterating features for player / surface / recency effects.

Real-Time Fractal Engine

[demos](#) | WebGL, Metal

- Mandelbrot, Julia, Newton, Kleinian—per-pixel fragment shaders with pan/zoom, supersampling, iteration coloring; plus a Swift/Metal Julia app for Apple GPUs.

Fourier Drawing Machine

[code](#) | [demo](#) | Python, JS

- Sampled paths → DFT coefficients → epicycle animation export; later moved the heavy work into a web worker for an interactive page.

Outside of code: marathon / triathlon training, calisthenics, squash.